

Hw03

[GhianiLaportMusmanno p.334]

Boscheim is a German company manufacturing electronics goods. Its KLR-12 CD player is specifically designed for the British market. KLR-12 is assembled in a plant near Rotterdam, then stocked in two warehouses located in Bristol and Middlesbrough and finally transported to the retailer outlets. The British market is divided into four sales districts whose centers of gravity are in London, Birmingham, Leeds and Edinburgh, and their yearly demands amount to 90000, 80000, 50000 and 70000 units, respectively.

The per-unit transport costs from assembly plant of Rotterdam to the warehouses of Bristol and Middlesbrough are 24.5 EUR and 26.0 EUR respectively, whereas the per-unit transport costs from the warehouses to the sale districts are reported in table.

Both warehouses have an estimated capacity of 15000 units and are supplied 10 times a year. Consequently, their maximum yearly throughput is 150000, i.e.,

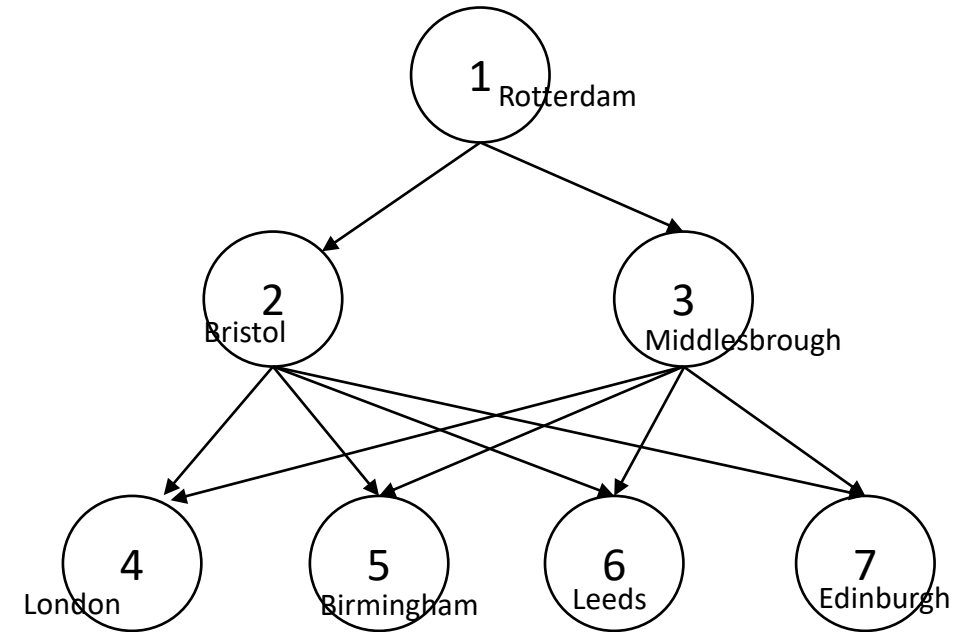
$$x_{12} \leq 150000$$

$$x_{13} \leq 150000$$

Assume that the capacity at the plant is equal to the total demand.

Write the mathematical programming model and solve for the best distribution plan.

The optimization can be solved using either Excel Solver or Python.



WH\Sales Districts	London	Birmingham	Leeds	Edinburgh
Bristol	9.6	7.0	15.2	28.5
Middlesbrough	19.5	13.3	5.0	11.3